

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination – 2023

Course: B. Tech. Branch: Computer Engineering Semester: Fifth

Subject Code & Name: BTCOC503 Software Engineering

Max Marks: 60 Date: 11 August 2023 Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. The use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) What is object-oriented programming (OOP)? Describe the fundamental principles of OOP and how they contribute to software design and development.	Understand	6
B) Provide an OO design for the following specifications. <i>“Build a software system that stores customers' contact details such as mobile number, WhatsApp number, mail ID and date of birth. The system should give birthday greetings and festival greetings to the customers.”</i> The OO design should include classes, relationships and methods. Draw a class diagram in UML.	Analyse	6
C) Compare and contrast agile and waterfall software development methodologies. Highlight the advantages and disadvantages of each approach.	Remember	6
Q.2 Solve Any Two of the following.		12
A) Define software testing and its importance in the development process. Describe at least three different types of testing and provide scenarios where each type would be most effective	Analyse	6
B) Explain the significance of continuous development and continuous integration in the DevOps project pipeline	Analyse	6
C) Suggest an appropriate process model for the projects. Justify your answers. 1. IOT-based water-level monitoring system. 2. Website development for a restaurant 3. A Mobile application to track students' attendance	Analyse	6
Q. 3 Solve Any Two of the following.		12

A)	Given a scenario, design a class diagram for a library management system that includes classes for books, library members, and transactions. Include associations, attributes, and methods.	Create	6
B)	Give the syntax of the following commands and explain their purpose in the context of GitHub. (1) fork, (2) merge, (3) pull request, (4) cloning a repository	Remember	6
C)	Consider a scenario where you are developing software for a medical clinic. Identify and categorize three functional and three non-functional requirements for the software. Explain why each requirement is important for the successful implementation of the system.	Remember	6
Q.4	Solve Any Two of the following.		12
A)	Imagine you're developing a banking application. Describe how you would handle concurrent access to a shared bank account to ensure data consistency and avoid conflicts.	Analyse	6
B)	Define the following quality attributes : (1) Modifiability, (2) Portability, (3) Scalability, (4) Reliability	Remember	6
C)	Analyse a recent software project failure and identify the key reasons behind its failure. Discuss what could have been done differently to prevent or mitigate these issues.	Remember	6
Q. 5	Solve the following.		12
A)	Write pseudocode for a function that takes a list of integers as input and returns the sum of all even numbers in the list.	Remember	6
B)	Define the following quality attributes : (1) Cohesion (2) Coupling (3) LOC (4) Fan-in and Fan-out	Create	6
C)	What is Hyrum's law? Explain with an example	Remember	6

*** End ***